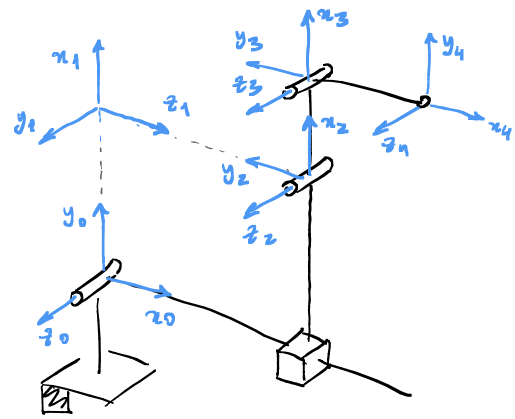
Sistemas Robóticos em Manipulação, E1 2021/2022 : Proposta de Solução A.CARVALHO

Note Title 2022-07-06

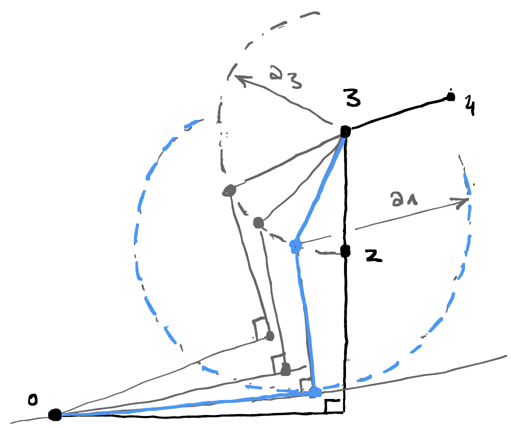
P1

(a)

d_i	θ_i	a_i	α_i
0	θ_1	a_1	$\pi/2$
d_2	0	0	$-\pi/2$
0	θ_3	a_3	0
0	θ_4	a_4	0



(b)



P2

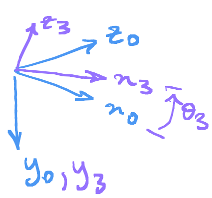
(a)

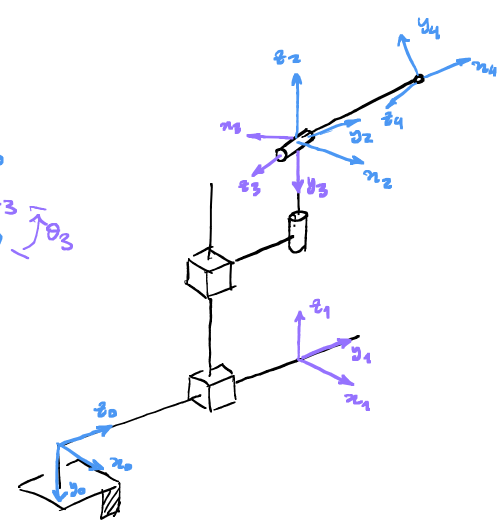
$$R^0_4 = R^0_3 R^3_4 ,$$

$$R^0_3 = (x\ y\ z)^0_3 = \begin{pmatrix} c_3 & 0 & -s_3 \\ 0 & 1 & 0 \\ s_3 & 0 & c_3 \end{pmatrix},$$

$$R^3_4 = R_z(\theta_4) = \begin{pmatrix} c_4 & -s_4 & 0 \\ s_4 & c_4 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

$$R^0_4 = \begin{pmatrix} c_3 c_4 & -c_3 s_4 & -s_3 \\ s_4 & c_4 & 0 \\ s_3 c_4 & -s_3 s_4 & c_3 \end{pmatrix}.$$





$$P_4 = d_1 z_0 + d_2 z_1 + a_4 x_4 = d_1 \hat{k} - d_2 \hat{j} + a_4 \begin{pmatrix} c_3 c_4 \\ s_4 \\ s_3 c_4 \end{pmatrix} = \begin{pmatrix} a_4 c_3 c_4 \\ -d_2 + a_4 s_4 \\ d_1 + a_4 s_3 c_4 \end{pmatrix}$$

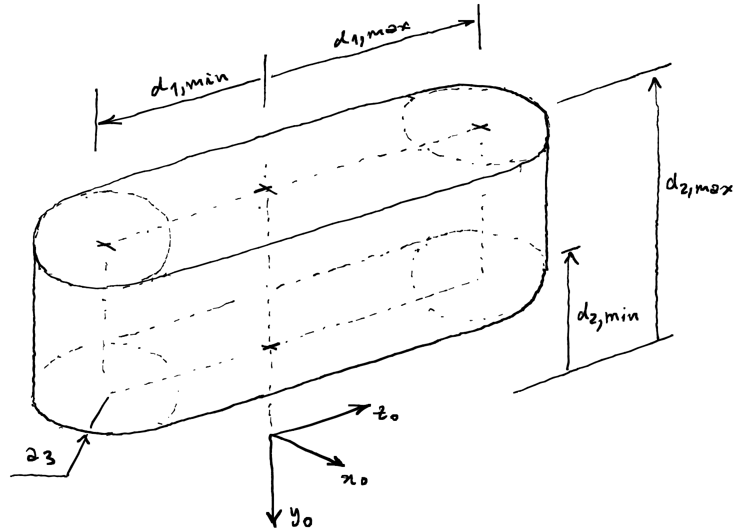
(b)

$$J_p = \frac{\partial P_4}{\partial q} = \left(\begin{array}{c|c|c|c} 0 & 0 & -a_4 s_3 c_4 & -a_4 c_3 s_4 \\ 0 & -1 & 0 & a_4 c_4 \\ 1 & 0 & a_4 c_3 c_4 & -a_4 s_3 s_4 \end{array}\right),$$

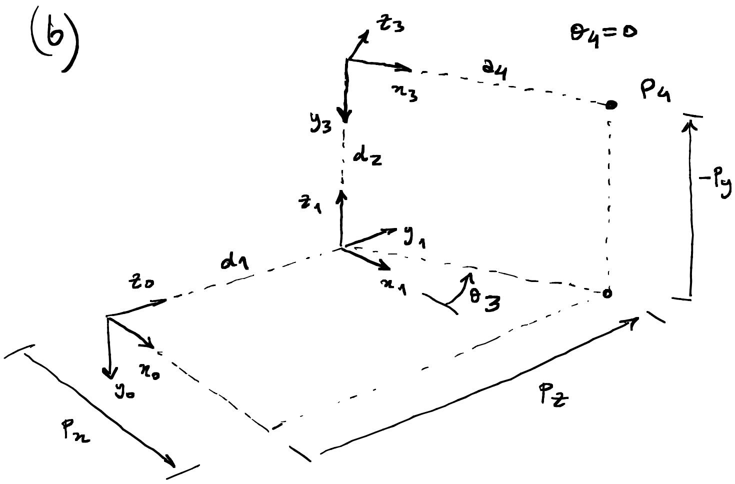
$$J_o = \left(0 \mid 0 \mid z_2 \mid z_3\right) = \left(\begin{array}{c|c|c|c} 0 & 0 & 0 & -s_3 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & c_3 \end{array}\right),$$

$$J = \begin{pmatrix} J_p \\ \hline J_o \end{pmatrix}.$$

P3 (a)



(b)



desired position: $(p_4)_d = (p_x \; p_y \; p_z)^T$

solution $(\sin\theta_3 > 0)$:

$$d_2 = -p_y \,.$$

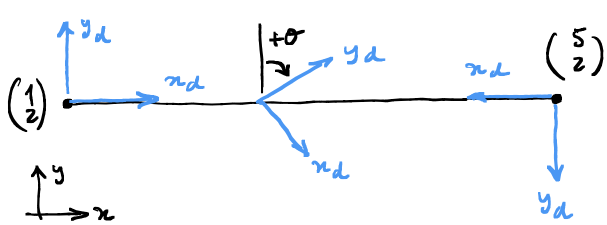
$$p_x = a_4 c_3 \Rightarrow c_3 = \frac{p_x}{a_4},$$

$$\Rightarrow \theta_3 = \arctan\left(\sqrt{1-c_3^2},\, c_3\right).$$

$$p_z = d_1 + a_4 s_3 \,,$$

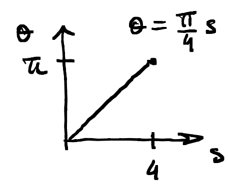
$$\Rightarrow d_1 = p_z - a_4 s_3 \,.$$

P4 (a)



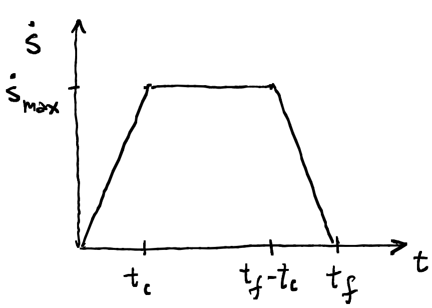
$$p(s) = \begin{pmatrix} 1+s \\ 2 \end{pmatrix}, \quad 0 \le s \le 4 \,.$$

$$R(\theta) = \begin{pmatrix} c_\theta & s_\theta \\ -s_\theta & c_\theta \end{pmatrix}, \quad 0 \le \theta \le \pi \,,$$



$$R(s) = \begin{pmatrix} \cos\left(\frac{\pi}{4}s\right) & \sin\left(\frac{\pi}{4}s\right) \\ -\sin\left(\frac{\pi}{4}s\right) & \cos\left(\frac{\pi}{4}s\right) \end{pmatrix}, \quad 0 \le s \le 4 \,.$$

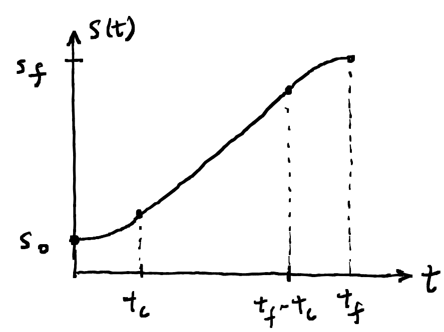
(b)



$$\dot{s}(t) = \begin{cases} \dfrac{\dot{s}_{max}}{t_c}\, t, & 0 \le t \le t_c \\ \dot{s}_{max}, & t_c < t \le t_f - t_c \\ -\dfrac{\dot{s}_{max}}{t_c}(t - t_f), & t_f - t_c < t \le t_f \end{cases}$$

(c)

$$s(t) = \int_0^t \dot{s}(u)\, du + s_0 \,.$$



$$s_f - s_0 = \int_0^{t_f} \dot{s}(u)\, du \overset{\text{Area}}{=} 2 \cdot \frac{\dot{s}_{max} \cdot t_c}{2} + \dot{s}_{max}(t_f - 2t_c)$$

$$= \dot{s}_{max}(t_f - t_c) \Rightarrow \dot{s}_{max} = \frac{s_f - s_0}{t_f - t_c} \,.$$

$$s_f - s_0 = 4 \Rightarrow \dot{s}_{max} = \frac{4}{t_f - t_c} \,.$$